```
clc;
clear;
close all;

% ============================================================
% IIR FILTER COEFFICIENTS
% ============================================================
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

N = 50;
x = [1 zeros(1,N-1)];       % Unit impulse input
n = 0:N-1;

% ============================================================
% 1. DIRECT FORM-I AND DIRECT FORM-II
% ============================================================
% MATLAB filter() computes the same transfer function
% using its standard implementation.
y1 = filter(b,a,x);         % Direct Form-I reference
y2 = filter(b,a,x);         % Direct Form-II reference

% ============================================================
% 2. CASCADE REALIZATION
% ============================================================
[sos,g] = tf2sos(b,a);

yc = g*x;

for k = 1:size(sos,1)
    yc = filter(sos(k,1:3),sos(k,4:6),yc);
end

% ============================================================
% 3. PARALLEL REALIZATION
% ============================================================
[r,p,k] = residuez(b,a);

% Direct polynomial term
yp = filter(k,1,x);

% Add first-order partial fractions
for i = 1:length(r)
    yp = yp + filter(r(i),[1 -p(i)],x);
end

% ============================================================
% 4. DISPLAY COEFFICIENTS
% ============================================================
disp('Numerator coefficients b =');
disp(b);
```

```
disp('Denominator coefficients a =');
disp(a);

disp('Second Order Sections (Cascade) =');
disp(sos);

disp('Cascade Gain =');
disp(g);

disp('Residues (Parallel) =');
disp(r);

disp('Poles (Parallel) =');
disp(p);

disp('Direct term =');
disp(k);

% ========================================================
% 5. ERROR AND PERCENTAGE DEVIATION
% ========================================================
E = [max(abs(y1-y2)); ...
     max(abs(y1-yc)); ...
     max(abs(y1-yp))];

reference = max(abs(y1));

if reference > 0
    D = E/reference*100;
else
    D = zeros(size(E));
end

Structure = {'DF-I vs DF-II'; ...
             'DF-I vs Cascade'; ...
             'DF-I vs Parallel'};

T = table(Structure,E,D, ...
    'VariableNames', ...
    {'Structure','Maximum_Error','Deviation_percent'});

disp('Comparison Table:');
disp(T);

% ========================================================
% 6. IMPULSE RESPONSE COMPARISON
% ========================================================
figure('Name','Impulse Response Comparison', ...
       'NumberTitle','off');

stem(n,y1,'filled');
hold on;
stem(n,y2,'--');
stem(n,yc,':','LineWidth',1.5);
```

```
stem(n,yp,'-.');

grid on;
xlabel('n');
ylabel('Amplitude');
title('Impulse Response of IIR Realizations');

legend('Direct Form-I','Direct Form-II', ...
       'Cascade','Parallel','Location','best');

hold off;

% ============================================================
% 7. NUMERICAL ERROR PLOT
% ============================================================
figure('Name','Numerical Error','NumberTitle','off');

plot(n,abs(y1-y2),'--','LineWidth',1.2);
hold on;
plot(n,abs(y1-yc),':','LineWidth',1.5);
plot(n,abs(y1-yp),'-.','LineWidth',1.2);

grid on;
xlabel('n');
ylabel('Absolute Error');
title('Numerical Error Between Realizations');

legend('DF-I vs DF-II', ...
       'DF-I vs Cascade', ...
       'DF-I vs Parallel', ...
       'Location','best');

hold off;

% ============================================================
% 8. FREQUENCY RESPONSE
% ============================================================
figure('Name','Frequency Response','NumberTitle','off');

freqz(b,a,1024);
title('Frequency Response of IIR Filter');

% ============================================================
% 9. POLE-ZERO PLOT
% ============================================================
figure('Name','Pole-Zero Plot','NumberTitle','off');

zplane(b,a);
title('Pole-Zero Plot of IIR Filter');
```

```
Numerator coefficients b =
    0.0048    0.0193    0.0289    0.0193    0.0048

Denominator coefficients a =
```

```
    1.0000   -2.3695    2.3140   -1.0547    0.1874

Second Order Sections (Cascade) =
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327

Cascade Gain =
    0.0048

Residues (Parallel) =
  -0.2698 + 0.1343i
  -0.2698 - 0.1343i
   0.2594 - 0.8024i
   0.2594 + 0.8024i

Poles (Parallel) =
   0.6604 + 0.4433i
   0.6604 - 0.4433i
   0.5243 + 0.1459i
   0.5243 - 0.1459i

Direct term =
    0.0256

Comparison Table:
         Structure          Maximum_Error    Deviation_percent
    ____________________    _____________    _________________

    {'DF-I vs DF-II'   }              0                  0
    {'DF-I vs Cascade' }     1.9221e-15          8.242e-13
    {'DF-I vs Parallel'}     1.9811e-15         8.4949e-13

Warning: Using only the real component of complex data.
```

Published with MATLAB® R2026b